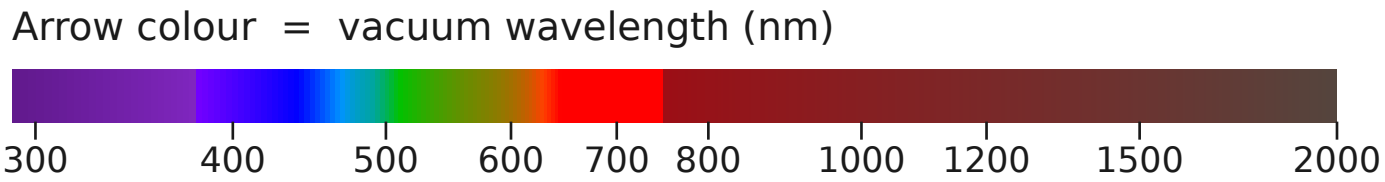


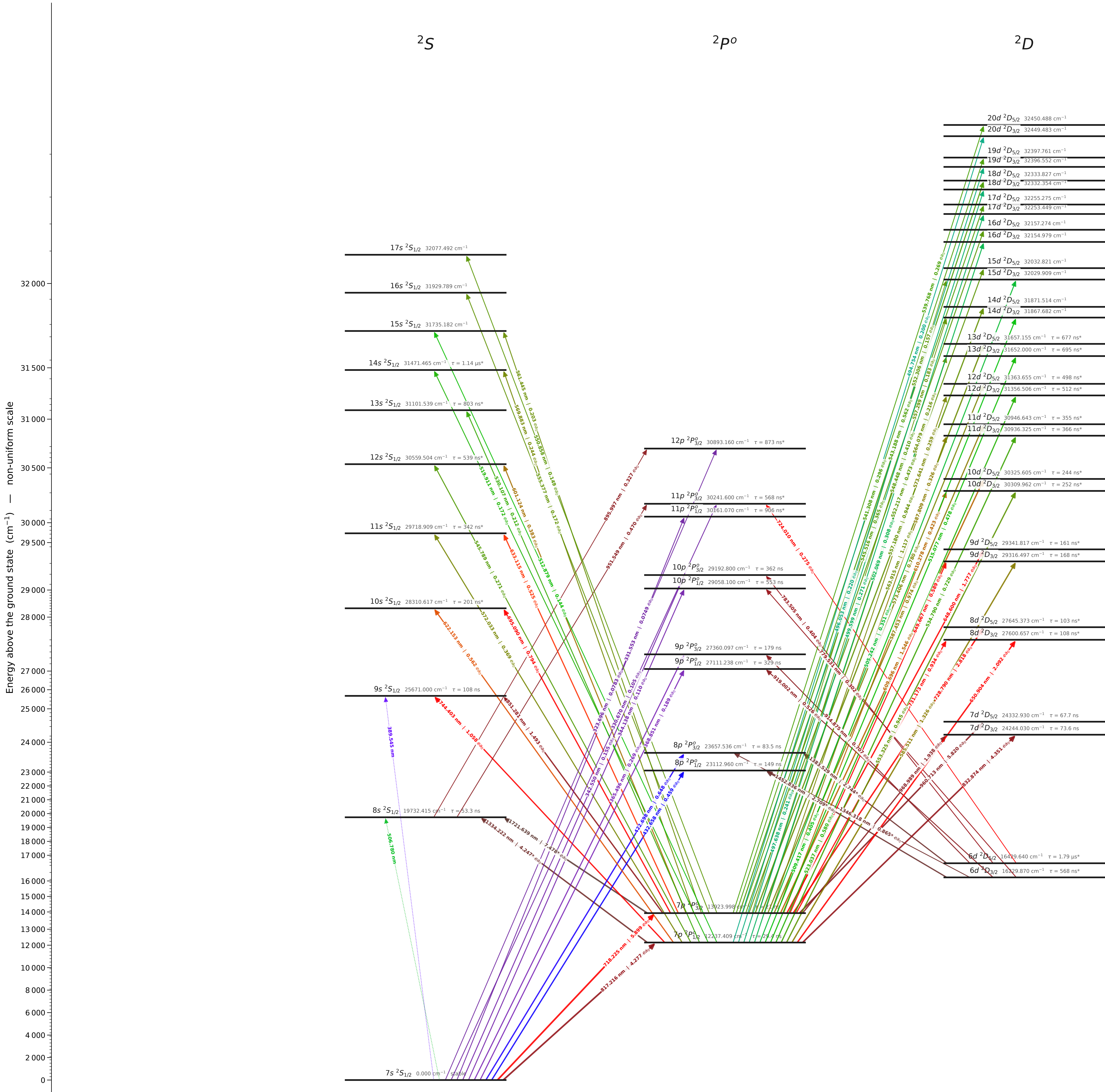
Francium

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||e\mathbf{r}||J'\rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Fr (Fr I). Strongest classified transitions below 2 μm (NIST): 52 levels, 77 lines.

Fr⁺ ionisation limit 32848.87 cm^{−1} = 4.07274 eV (304.424 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e\mathbf{r} J'\rangle$ ($e a_0$)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
<i>7s</i> ² <i>S</i> _{1/2} – <i>9s</i> ² <i>S</i> _{1/2}	389.5446	389.4343	769.5972	–	–	1.48	–	two-photon 75-9S spectroscopy of trapped 210Fr
<i>7s</i> ² <i>S</i> _{1/2} – <i>8p</i> ² <i>P</i> _{3/2}	422.6983	422.5793	709.23507(3)	0.648	2.820e+06	1.906	45.3	blue line
<i>7s</i> ² <i>S</i> _{1/2} – <i>8p</i> ² <i>P</i> _{1/2}	432.6577	432.5361	692.9091	0.459	2.640e+06	1.066	17.4	blue line
<i>7s</i> ² <i>S</i> _{1/2} – <i>8s</i> ² <i>S</i> _{1/2}	506.7803	506.6391	591.5629	–	–	2.986	–	two-photon / parity-violation transition 7S-8S
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.2251	718.0272	417.40739(6)	5.899	4.757e+07	7.572	100	D2 line
<i>7p</i> ² <i>P</i> _{1/2} – <i>9s</i> ² <i>S</i> _{1/2}	744.4026	744.1976	402.7289	1.058	2.750e+06	1.48	27.2	744.4 nm: decay channel of 9S
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{1/2}	817.2165	816.9918	366.84584(6)	4.277	3.396e+07	5.404	100	D1 line
<i>7p</i> ² <i>P</i> _{3/2} – <i>7d</i> ² <i>D</i> _{3/2}	832.8738	832.6449	359.9494	4.351	1.660e+07	2.162	78.8	832.9 nm: 7P1/2 - 7D3/2
<i>7p</i> ² <i>P</i> _{3/2} – <i>9s</i> ² <i>S</i> _{1/2}	851.2810	851.0472	352.1663	1.493	3.660e+06	1.48	34.1	851.3 nm: 7P3/2 - 9S step
<i>7p</i> ² <i>P</i> _{3/2} – <i>7d</i> ² <i>D</i> _{5/2}	960.7134	960.4499	312.0519	5.820	1.290e+07	2.351	99.5	960.7 nm: 7P3/2 - 7D5/2
<i>7p</i> ² <i>P</i> _{3/2} – <i>7d</i> ² <i>D</i> _{3/2}	968.9892	968.7236	309.3868	1.938	2.090e+06	2.162	19.3	969.0 nm: 7P3/2 - 7D3/2
<i>7p</i> ² <i>P</i> _{3/2} – <i>8s</i> ² <i>S</i> _{1/2}	1334.2218	1333.8570	224.6946	4.247*	7.693e+06	2.986	40.9	1.33 um: second step of the two-photon 7S-8S excitation u...
<i>7p</i> ² <i>P</i> _{3/2} – <i>8s</i> ² <i>S</i> _{1/2}	1721.6395	1721.1694	174.1320	7.478*	1.110e+07	2.986	59.1	1.72 um: decay channel of 8S

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²¹²Fr hyperfine structure (I = 5)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
<i>7s</i> ² <i>S</i> _{1/2}	9064.20(20)	–	F=9/2: -27192.60 F=11/2: +22660.50
<i>7p</i> ² <i>P</i> _{1/2}	1192.00(20)	–	F=9/2: -3576.00 F=11/2: +2980.00
<i>7p</i> ² <i>P</i> _{3/2}	97.21(10)	-26.00(20)	F=7/2: -886.16 F=9/2: -431.81 F=11/2: +107.61 F=13/2: +722.58
<i>8p</i> ² <i>P</i> _{3/2}	373.00(10)	–	F=9/2: -1119.00 F=11/2: +932.50
<i>8p</i> ² <i>P</i> _{3/2}	32.80(10)	-7.7(9)	F=7/2: -298.54 F=9/2: -145.93 F=11/2: +35.88 F=13/2: +244.07
<i>8d</i> ² <i>D</i> _{3/2}	13.0(6)	–	F=7/2: -117.00 F=9/2: -58.50 F=11/2: +13.00 F=13/2: +97.50
<i>8d</i> ² <i>D</i> _{5/2}	-7.2(6)	–	F=5/2: +108.00 F=7/2: +82.80 F=9/2: +50.40 F=11/2: +10.80 F=13/2: -36.00 F=15/2: -90.00
<i>10s</i> ² <i>S</i> _{1/2}	401(5)	–	F=9/2: -1203.00 F=11/2: +1002.50
<i>9d</i> ² <i>D</i> _{3/2}	7.1(7)	–	F=7/2: -63.90 F=9/2: -31.95 F=11/2: +7.10 F=13/2: +53.25
<i>9d</i> ² <i>D</i> _{5/2}	-3.6(4)	–	F=5/2: +54.00 F=7/2: +41.40 F=9/2: +25.20 F=11/2: +5.40 F=13/2: -18.00 F=15/2: -45.00
<i>11s</i> ² <i>S</i> _{1/2}	225(3)	–	F=9/2: -675.00 F=11/2: +562.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
<i>7s</i> ² <i>S</i> _{1/2} – <i>8p</i> ² <i>P</i> _{3/2}	422.698	213-212	-1615.0(3)
<i>7s</i> ² <i>S</i> _{1/2} – <i>8p</i> ² <i>P</i> _{3/2}	422.698	220-212	-20564.0(7)
<i>7s</i> ² <i>S</i> _{1/2} – <i>8p</i> ² <i>P</i> _{3/2}	422.698	221-212	-23298.0(8)
<i>7s</i> ² <i>S</i> _{1/2} – <i>8s</i> ² <i>S</i> _{1/2}	506.780	213-208	-5123(6)
<i>7s</i> ² <i>S</i> _{1/2} – <i>8s</i> ² <i>S</i> _{1/2}	506.780	213-209	-3678(6)
<i>7s</i> ² <i>S</i> _{1/2} – <i>8s</i> ² <i>S</i> _{1/2}	506.780	213-210	-3274(6)
<i>7s</i> ² <i>S</i> _{1/2} – <i>8s</i> ² <i>S</i> _{1/2}	506.780	213-211	-1958(6)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	207-212	5239(4)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	208-212	5003(3)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	209-212	3133.0(20)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	210-212	2603.0(10)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	211-212	901(3)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	222-212	-26262(3)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	223-212	-27922.0(20)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	224-212	-30891.0(10)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	225-212	-32297(6)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	226-212	-34401.0(10)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	227-212	-38352.0(20)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	228-212	-40077(5)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	221-212	-23570.0(20)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	213-212	-1641(3)
<i>7s</i> ² <i>S</i> _{1/2} – <i>7p</i> ² <i>P</i> _{3/2}	718.225	220-212	-20806.8(5)

Shift = ν(first isotope) – ν(second isotope).